



MRC Cognition
and Brain
Sciences Unit



UNIVERSITY OF
CAMBRIDGE

EEG/MEG 3:

Time-Frequency Analysis

Olaf Hauk

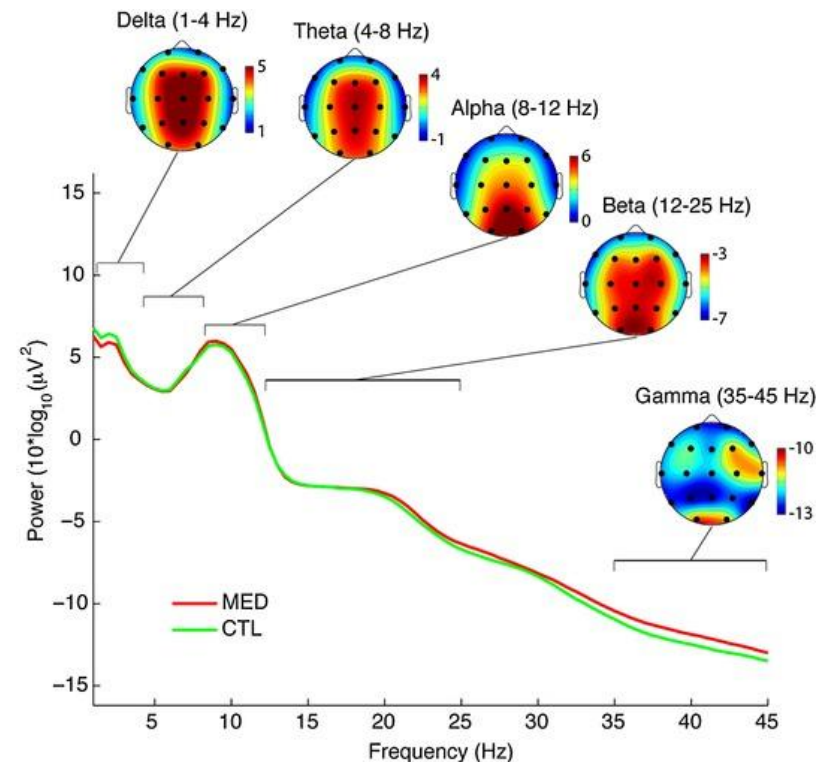
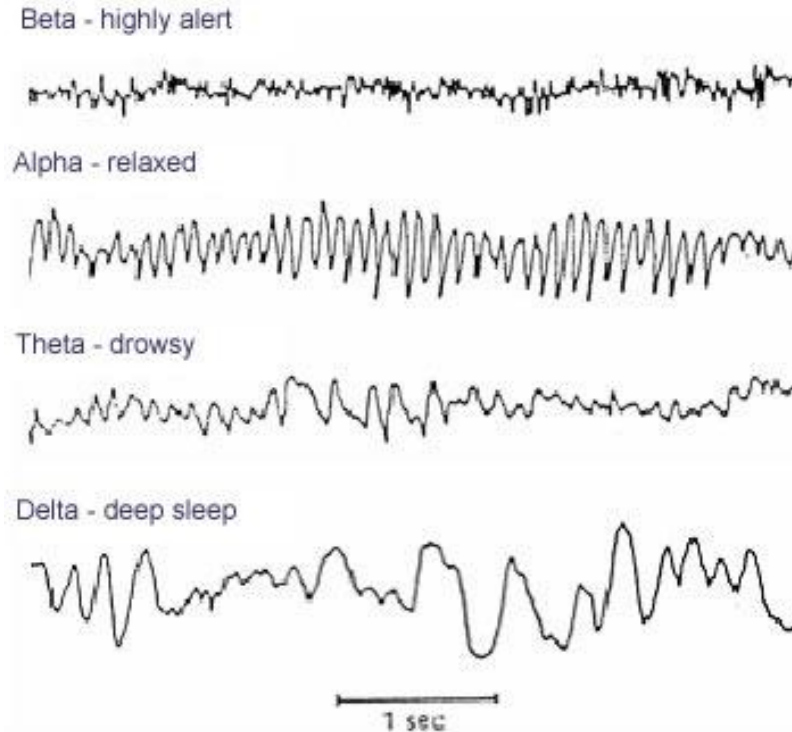
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“Brain Rhythms” and “Oscillations”

Time course and topography may differ among different frequency bands

(and may depend on task, environment, subject group etc.)

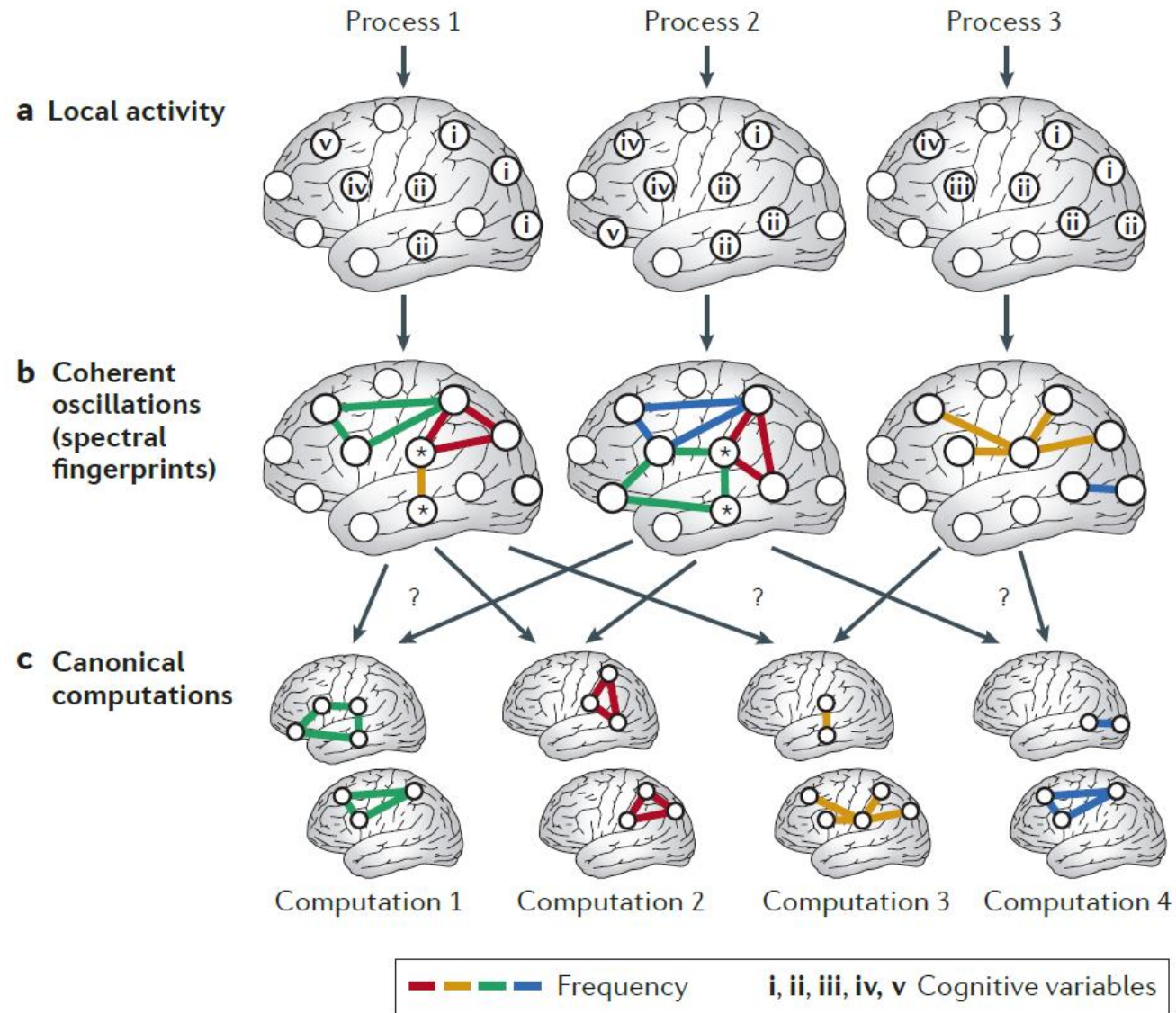
=> Different frequency “bands” may reflect different processes/computations, systems/networks, etc.



Cahn et al., Cogn Proc 2010,

<http://link.springer.com/article/10.1007%2Fs10339-009-0352-1/>

“Brain Rhythms” and “Oscillations”



Polar Representation Of Periodic Signals

Euler's Formula

“**Complex**” numbers can capture the two axes of the coordinate system for the circle around which the vector rotates periodically – this is rather abstract but helps the notation enormously.

$$e^{-i\theta} = \cos(\theta) + i * \sin(\theta) \quad i = \sqrt{-1}$$

Therefore:

$$\cos(\theta) = \text{real}(e^{-i\theta})$$

$$\sin(\theta) = \text{imag}(e^{-i\theta})$$

An oscillation at a particular frequency can be described in a “polar representation”:

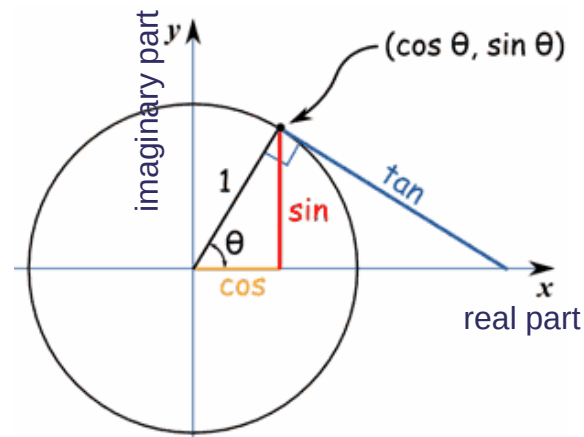
$$a * e^{-i2\pi ft}$$

a : amplitude

2π : circumference of unit circle

f : frequency

t : time



The Polar Representation Of Periodic Signals

Convenient To Compare Periodic Signals

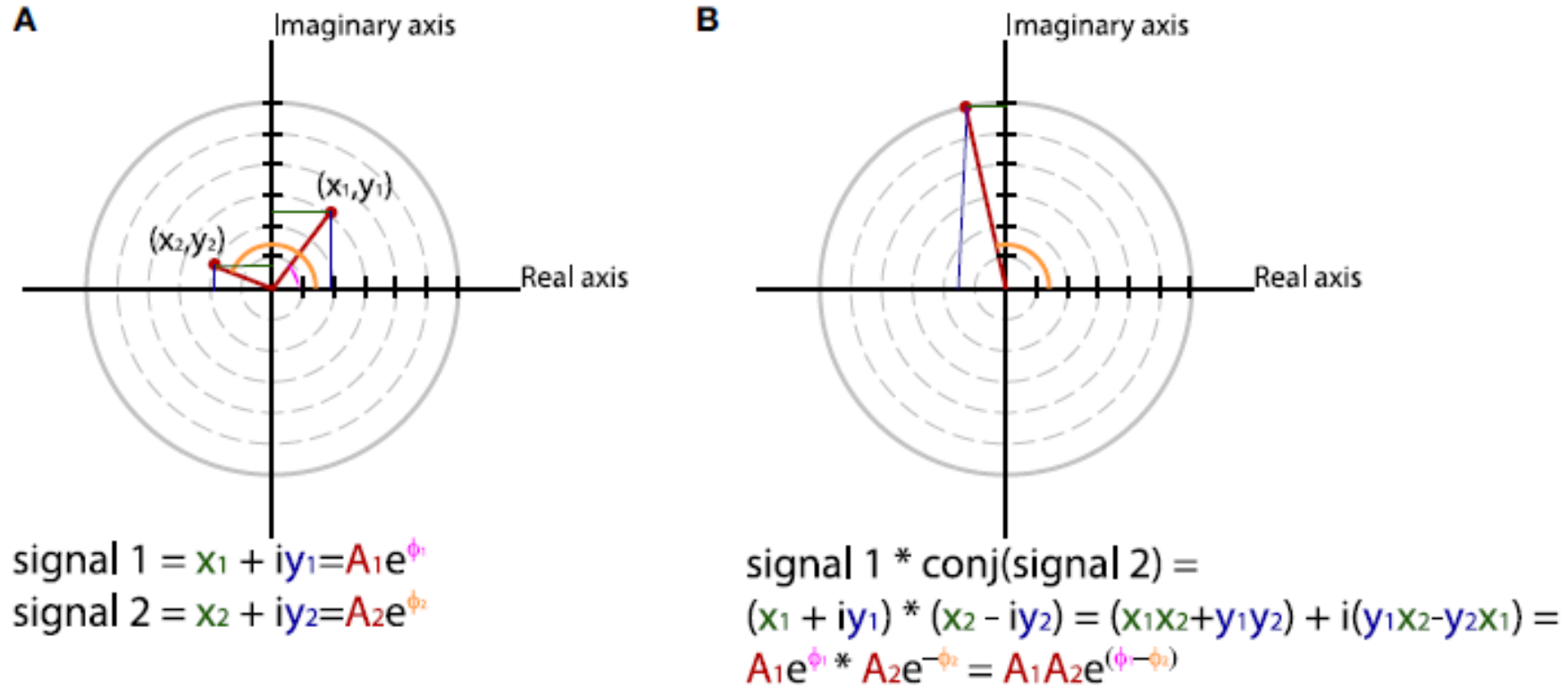
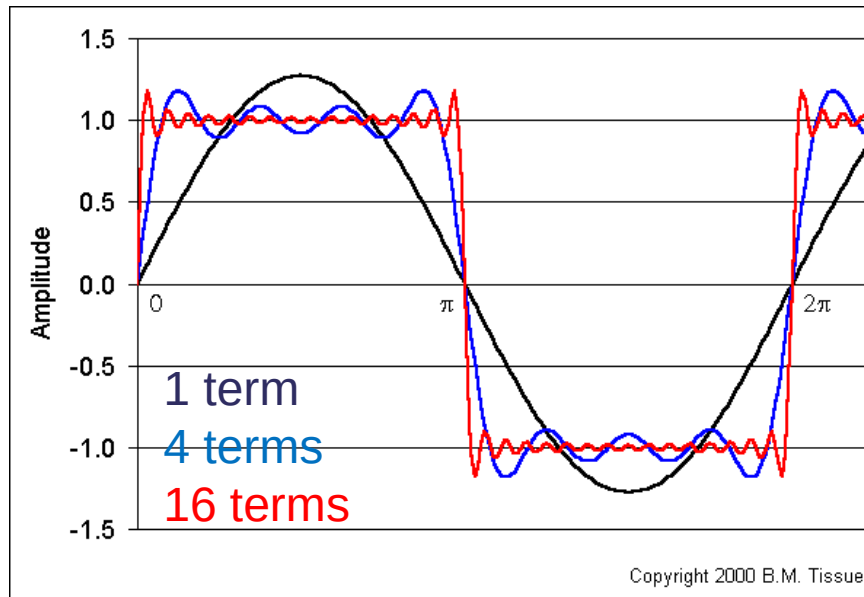


FIGURE 2 | Using polar coordinates and complex numbers to represent signals in the frequency domain. (A) The phase and amplitude of two signals. **(B)** The cross-spectrum between signal 1 and 2, which corresponds to multiplying the amplitudes of the two signals and subtracting their phases.

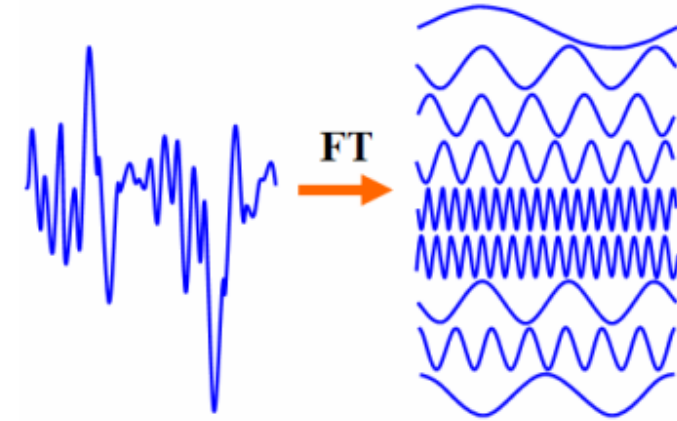
The Fourier Decomposition

The Fourier (De-)Composition

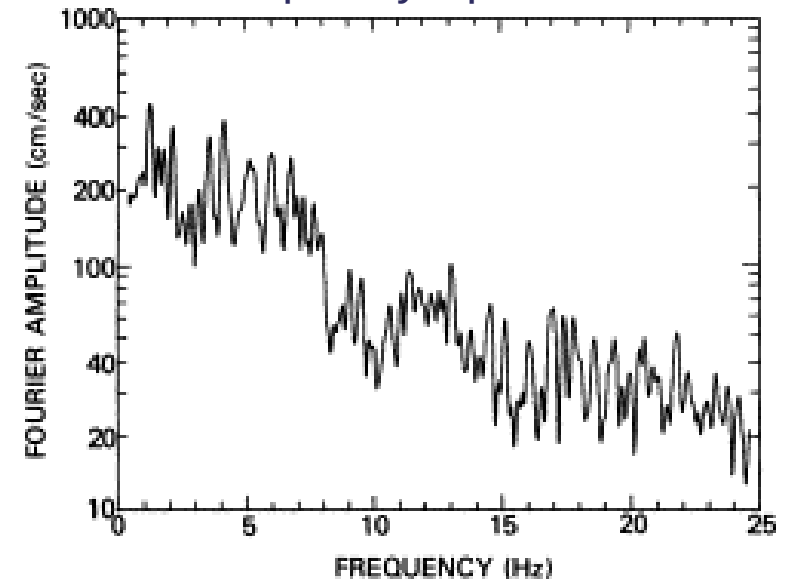
Approximating a step function
with Fourier terms



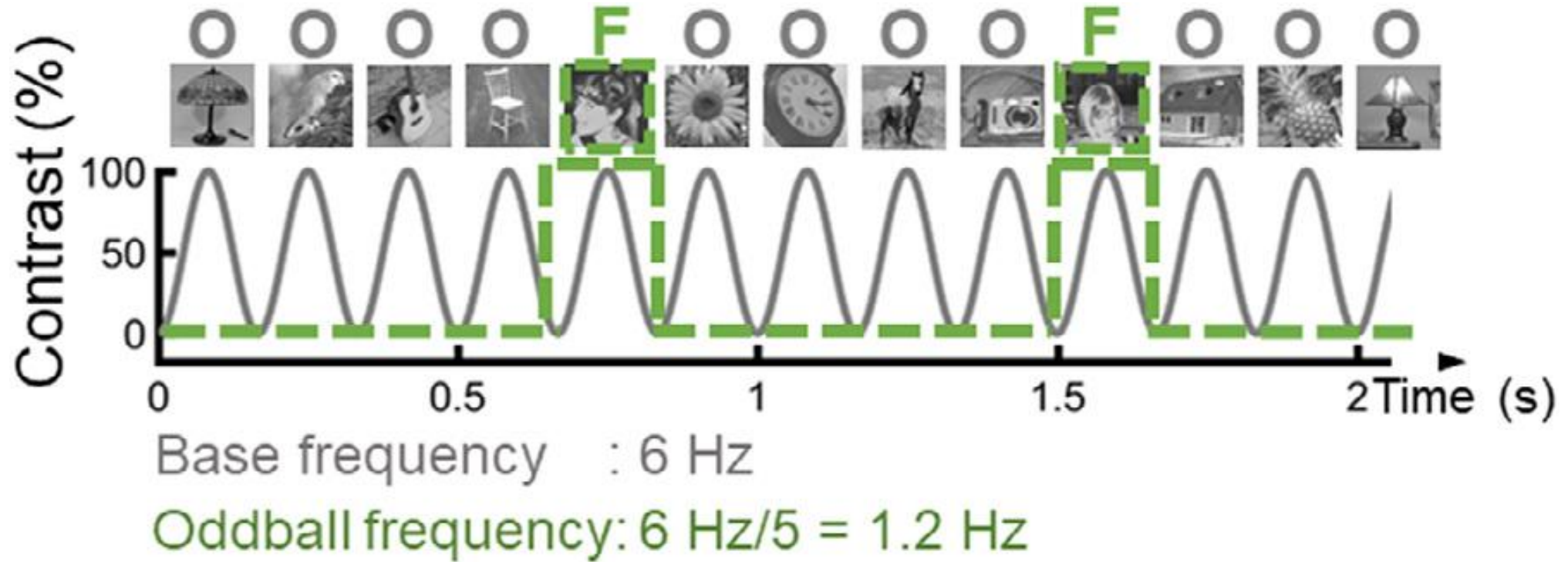
Decomposing signals
into sine/cosine terms



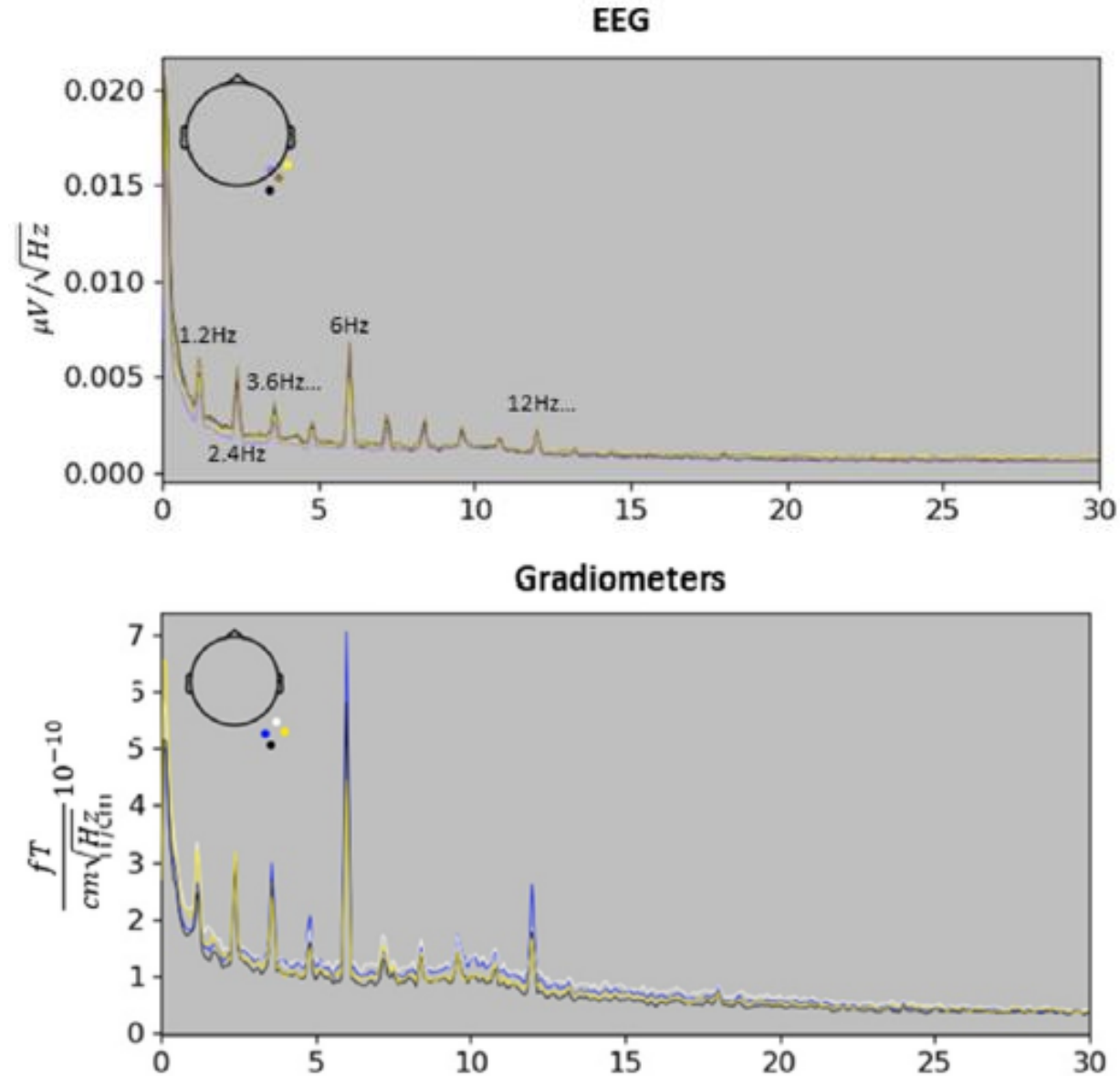
Frequency Spectrum



Example: Fast Periodic Visual Stimulation (FPVS)

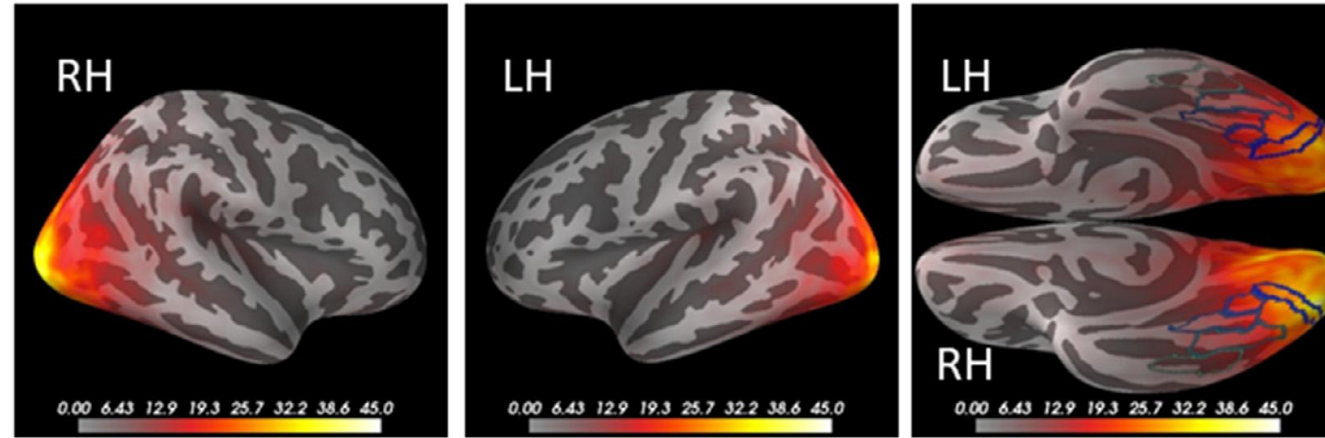


Fast Periodic Visual Stimulation (FPVS)

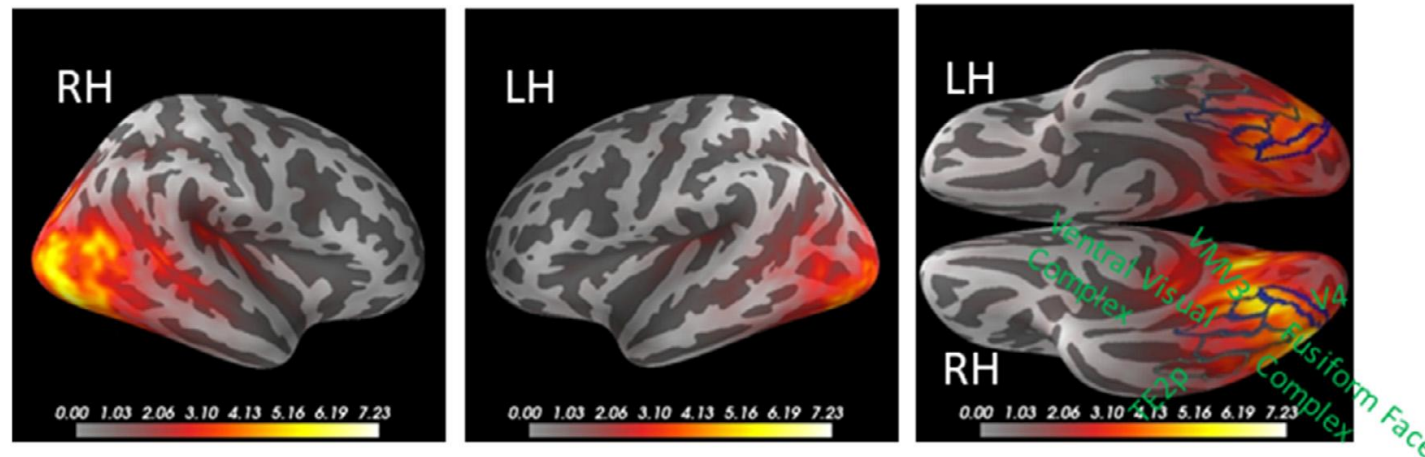


Fast Periodic Visual Stimulation (FPVS)

Base Frequency



Face-selective Frequency

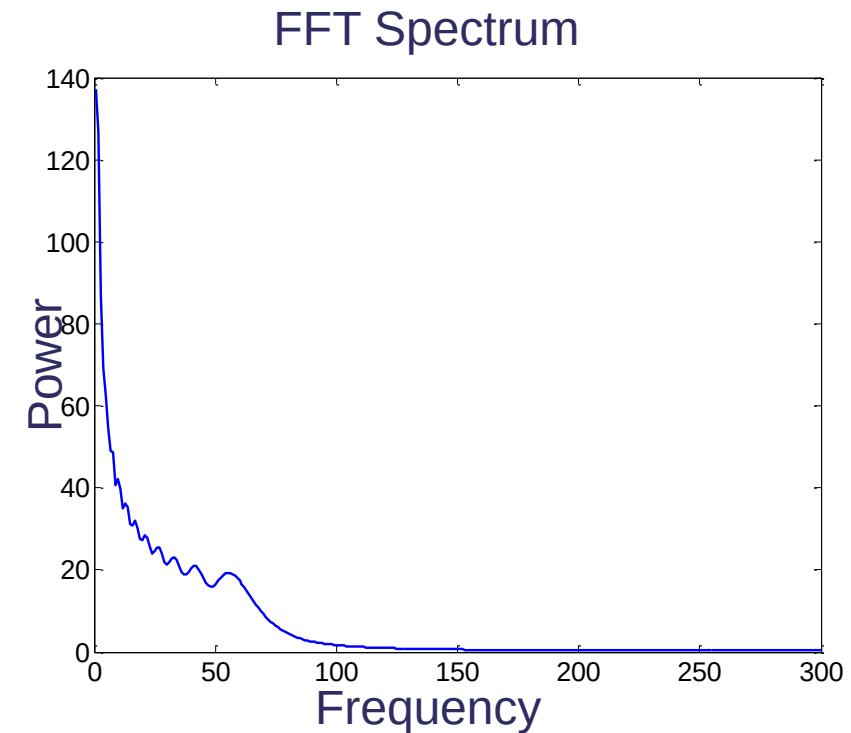
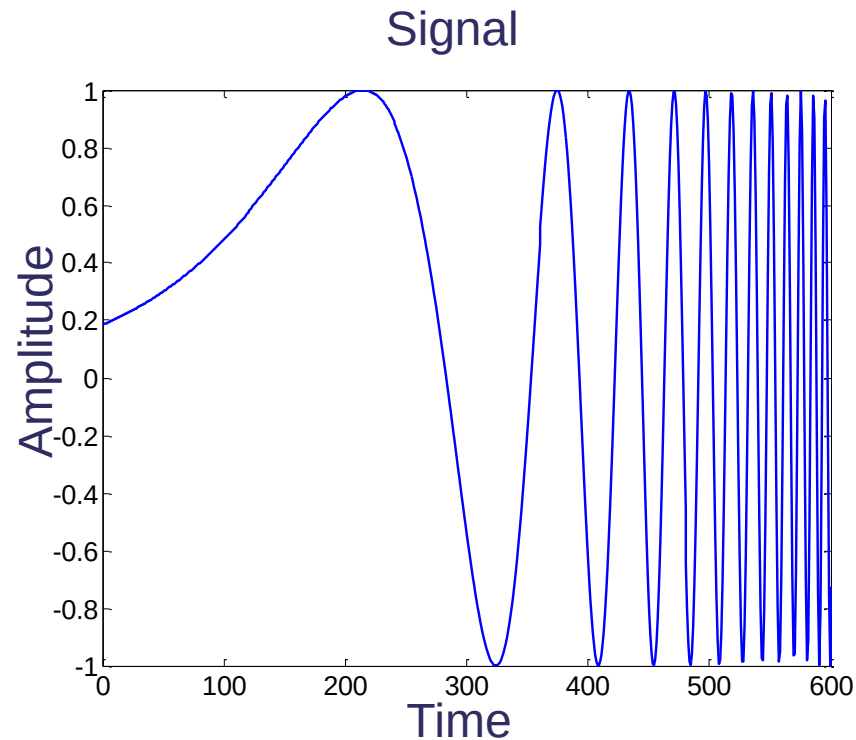


Time-Frequency Analysis

Motivation for Time-Frequency Analysis

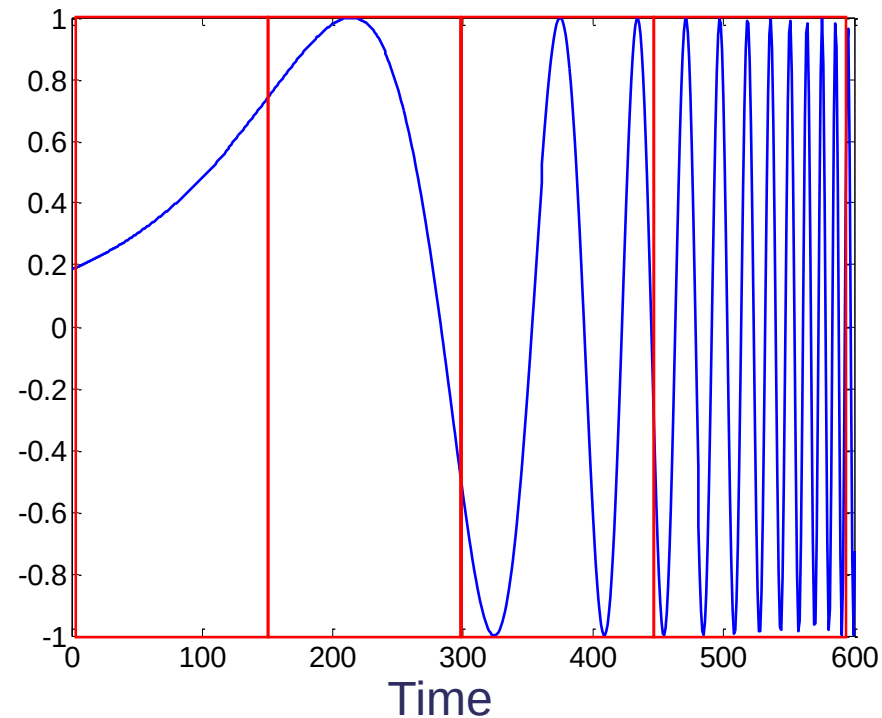
Fourier Transform assumes sines and cosines with constant amplitudes across the whole time series (“stationarity”).

What are we going to do with signals where activity in frequency bands changes over time, e.g. with a signal like this?



Motivation for Time-Frequency Analysis

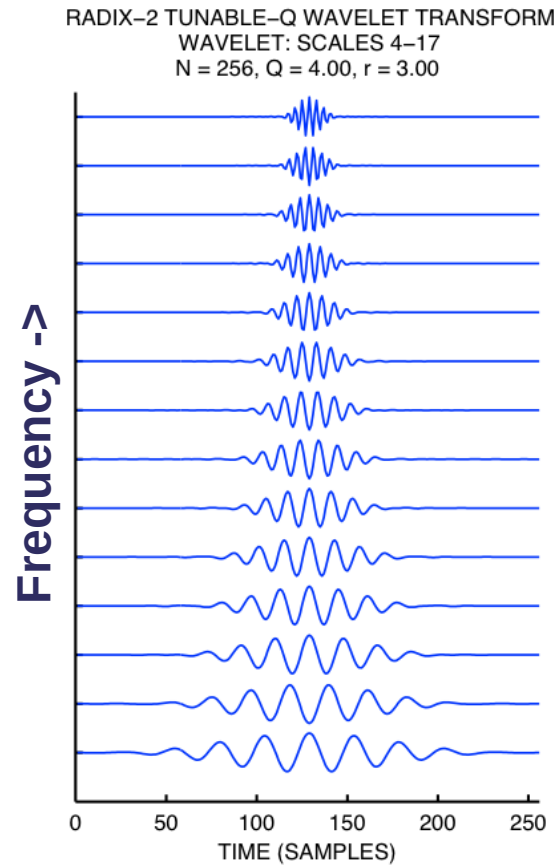
You could run separate FFTs for different (sliding) time windows:



But different window sizes are more or less optimal for different frequencies.
Run different FFTs with different window sizes for different frequency ranges? Ouff.

Time-Frequency Analysis: Wavelets (“little waves”)

Wavelets provide an optimal trade-off between frequency and time resolution.



Wavelets are getting
“broader” with
decreasing frequency

=>

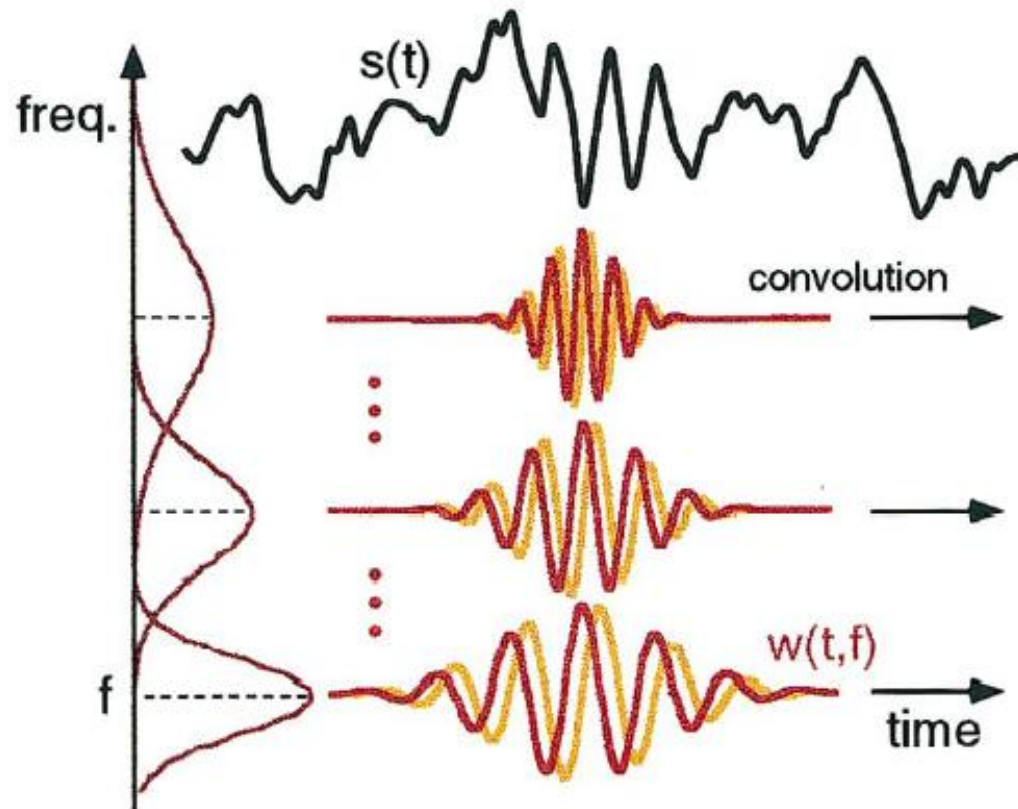
Time resolution
decreases as
frequency decreases

Wavelets are convolved with the data to give instantaneous amplitude and phase estimates for different frequency ranges.

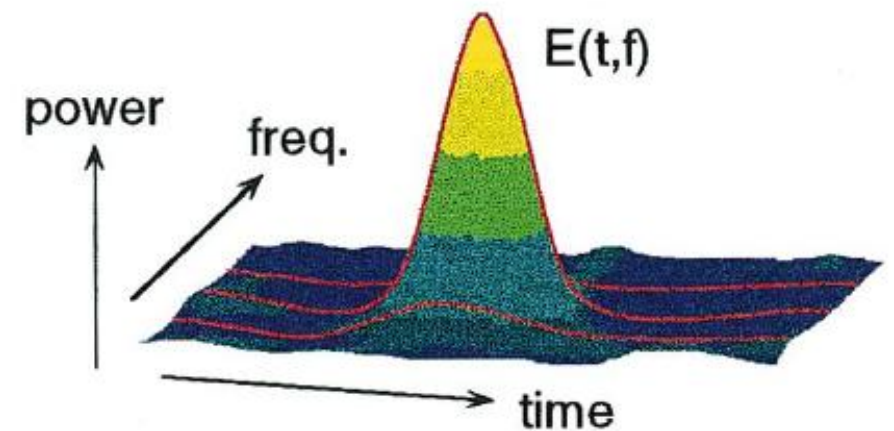
Time-Frequency Analysis: Wavelets

Wavelet Transform

Trade-off between time and frequency resolution



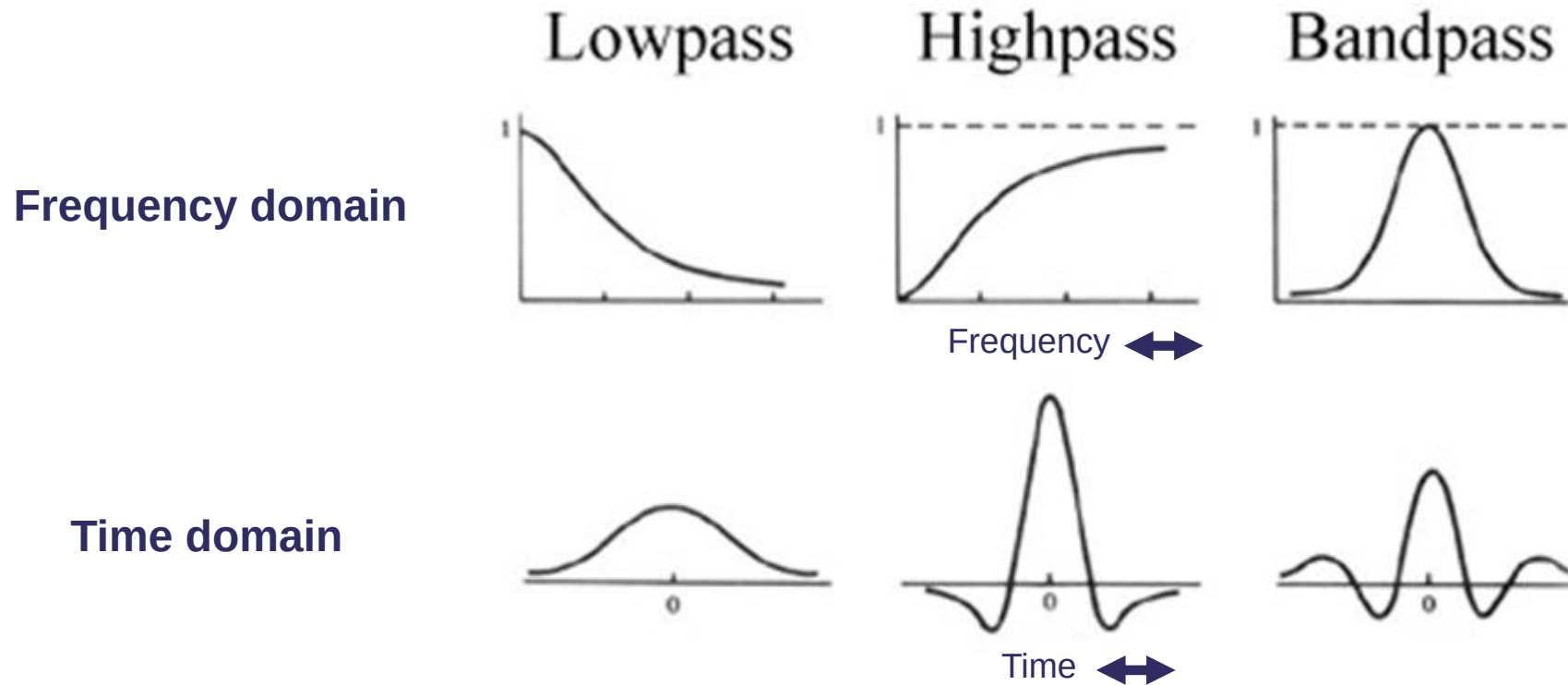
Time-Frequency Power



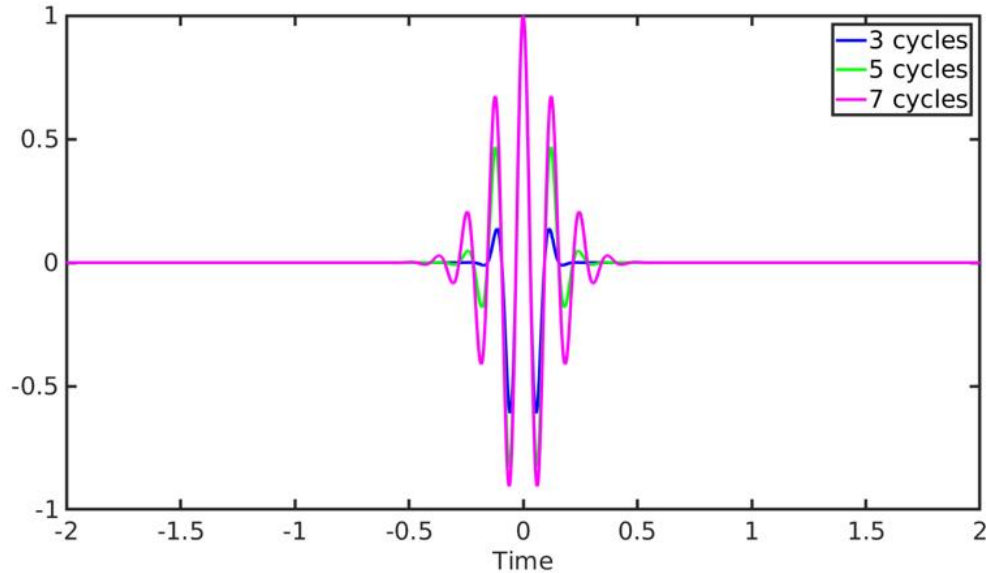
Basic Principals of Frequency Filtering

Time-domain and frequency-domain filtering are two sides of the same coin:

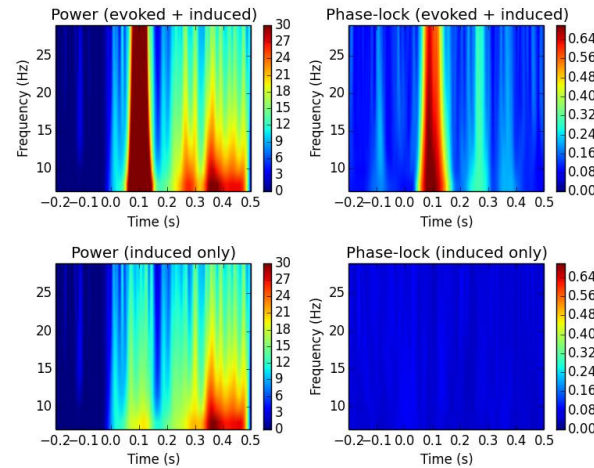
One type of frequency-domain filtering corresponds to one type of time-domain filtering.



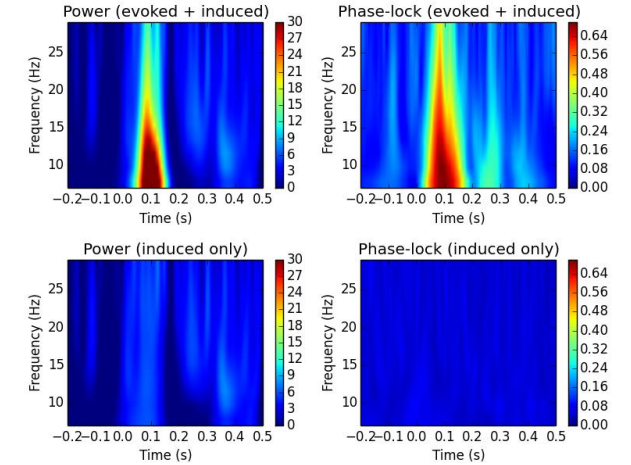
Effect of Number of Cycles



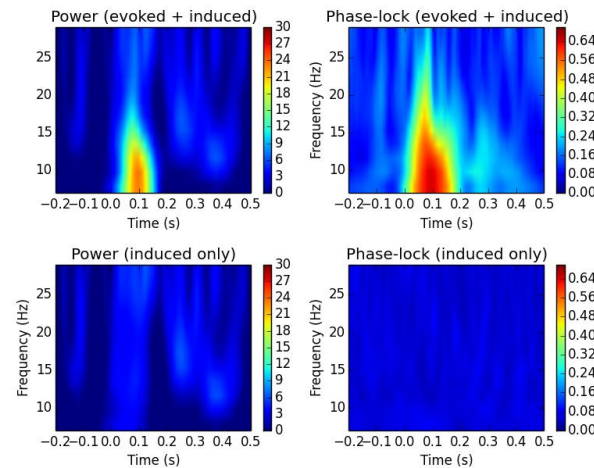
1 cycle



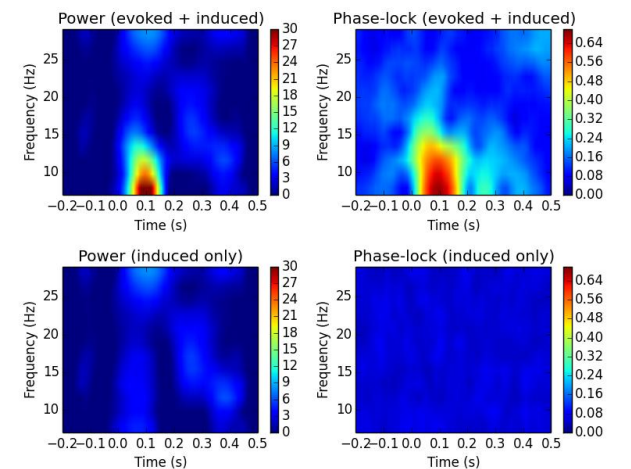
2 cycles



3 cycles

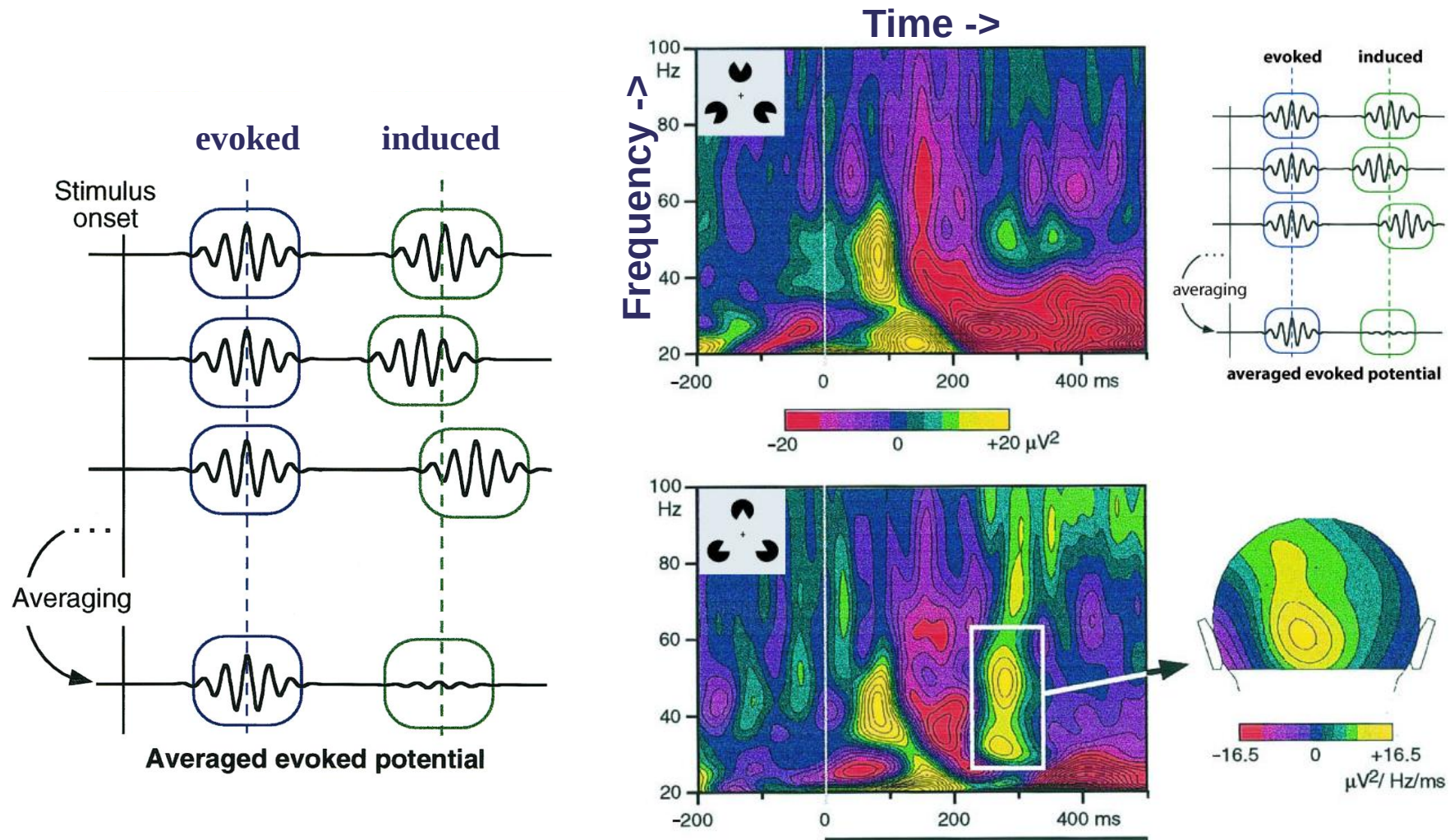


Freq/3 cycles



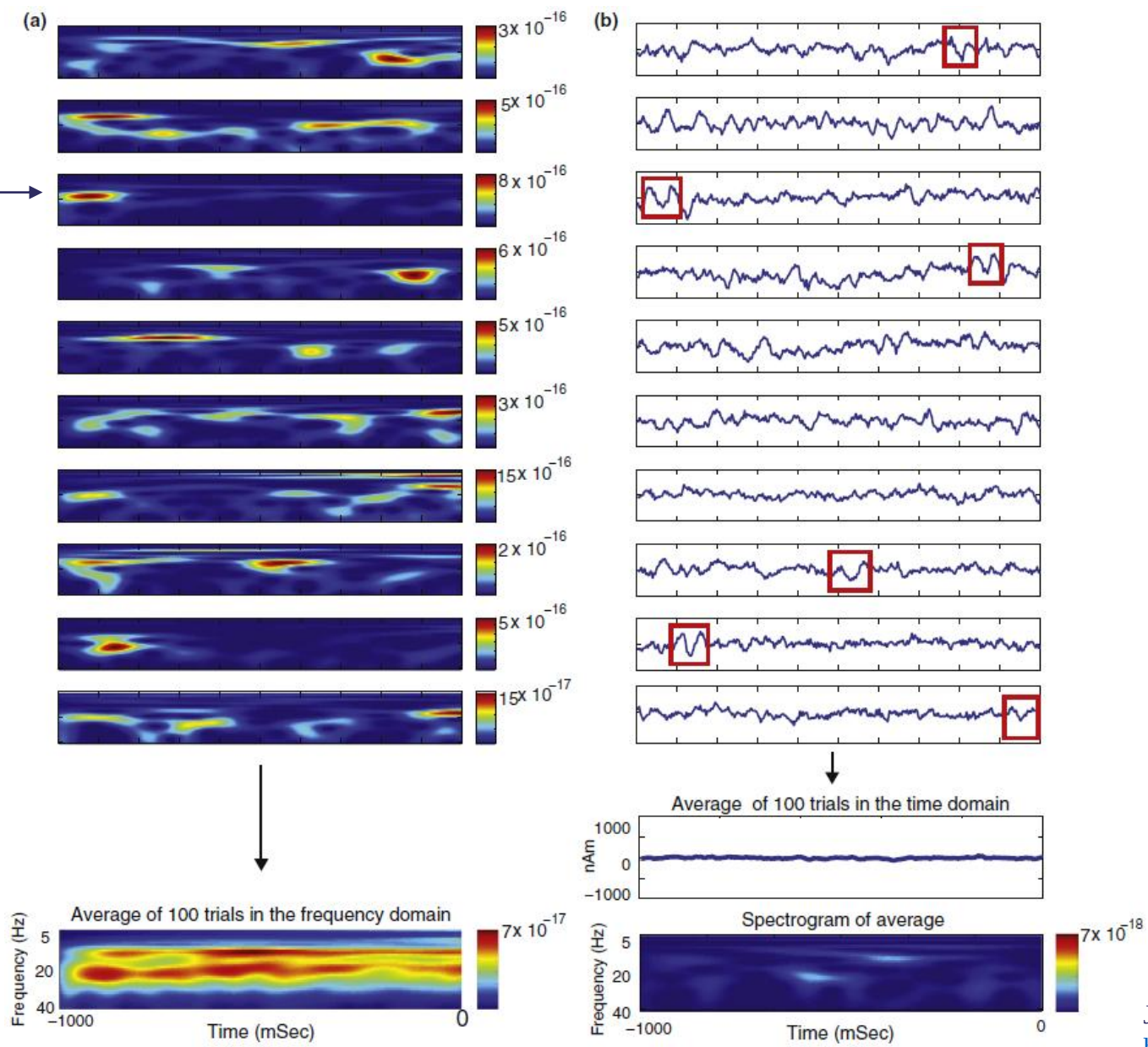
Rule of thumb: For low frequencies ($< \sim 10\text{Hz}$), $n=2$ or 3 ; for higher frequencies $n=f/3$.

Evoked and Induced Rhythmic Activity



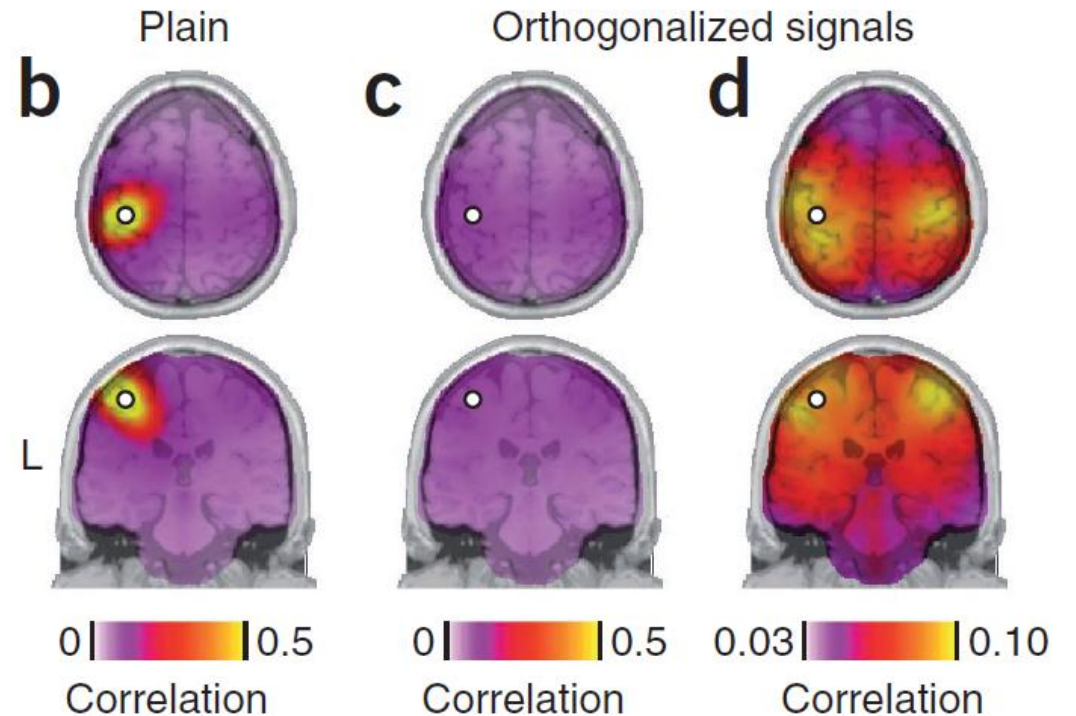
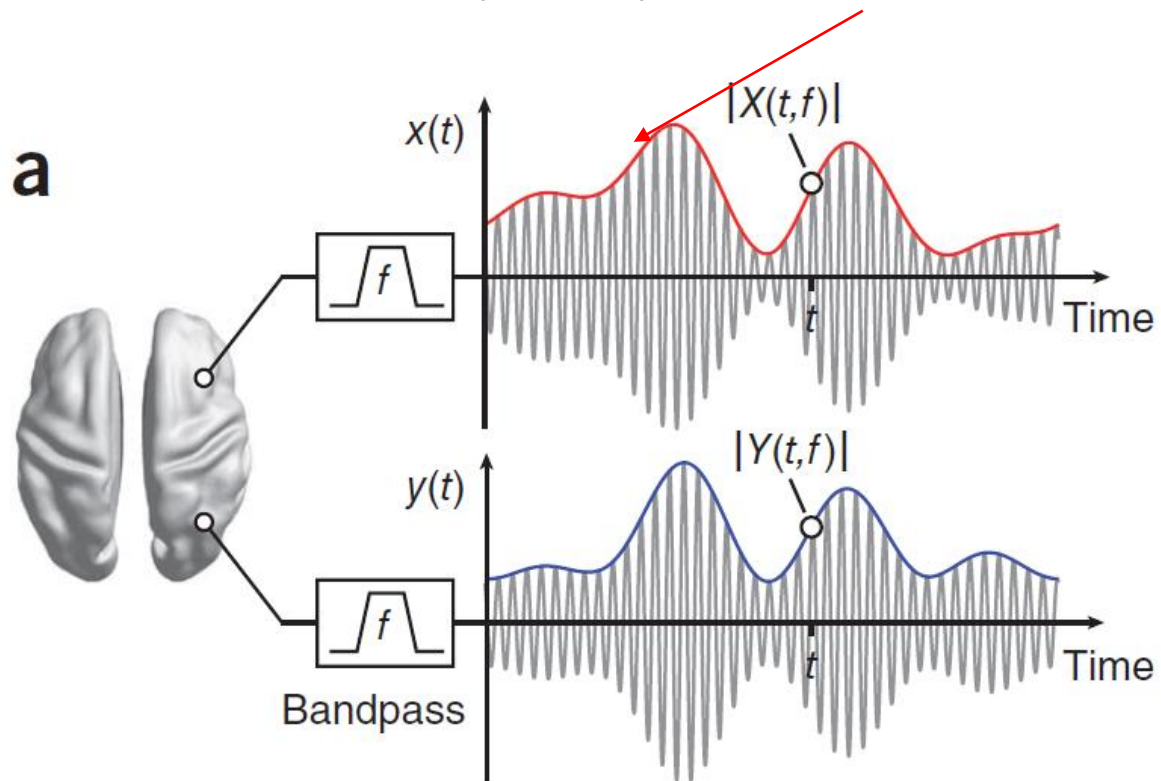
When brain rhythms aren't “rhythmic” – the example of beta “oscillations”

“beta bursts”
rather than “oscillations”



Alternative to wavelets: Hilbert Transform

(“Hilbert”) Envelope for a frequency band





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Thank you